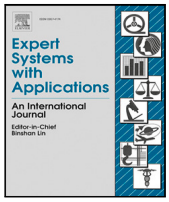
Expert Systems With Applications 238 (2024) 122391



Contents lists available at ScienceDirect

Expert Systems With Applications

journal homepage: www.elsevier.com/locate/eswa





An attention-based CNN for automatic whole-body postural assessment

Zewei Ding [a], Wanqing Li [a,*], Jie Yang [a], Philip Ogunbona [a], Ling Qin [b]

[a] Advanced Multimedia Research Lab, University of Wollongong, Australia
[b] The People's Hospital of Guangxi Zhuang Autonomous Region, Nanning, China

ARTICLE INFO

Keywords:
Postural assessment
Convolutional Neural Network (CNN)
Rapid Entire Body Assessment (REBA)
Attention
Musculoskeletal Disorder (MSD)

ABSTRACT

Fully automatic postural assessment is highly useful, but has been challenging. Conventional methods either require manual assessment by ergonomists or depend on special devices that are intrusive, thus being hardly feasible in daily activities and workplaces. In this work, an attention-based convolutional neural network (CNN) is developed for automatic whole-body postural assessment. The proposed network learns to identify highly relevant regions (or body parts) and extract features automatically. Risk of the posture is estimated from the extracted features accordingly. To evaluate the proposed method, a postural dataset, referred to as pH36M, is created by re-targeting Human3.6M, one of the largest publicly available datasets for pose estimation using the Rapid Entire Body Assessment (REBA) criteria. Experimental results on pH36M demonstrate that proposed method achieves promising performance in comparison to baselines and the average assessment scores are substantially aligned with human assessment with a Kappa value of 0.73.

1. Introduction

Postural assessment aims at identifying risk factors such as non-neutral postures, repetitive motions, and rapid build-up of force, providing a way to prevent the development of musculoskeletal disorders (MSDs). Ergonomists have developed a set of criteria, such as Rapid Entire Body Assessment (REBA) (Hignett & McAtamney, 2000) and Rapid Upper Limb Assessment (RULA) (McAtamney & Corlett, 1993), and observation based assessment tools/methods accordingly. In those methods, a scoring sheet is predefined using the criteria, for which raters/observers analyze the postures and further record results via direct observation or watching videos. There are, clearly, several drawbacks of these methods. First, the observation procedure is time consuming and labor intensive, thus making the methods unsuitable for the long-term and large-scale assessment. Second, the results suffer from both intra- and inter-rater variances, i.e. assessment highly depends on the rater's observing viewpoints and subject to their experience/preference. Additionally, privacy is often a major concern.

Recently, automatic methods have been proposed to identify risks directly from images and/or videos using machine/deep learning and computer vision techniques without involving professional raters/observers (Ding, Li, Ogunbona, & Qin, 2019; Li, Martin, & Xu, 2020; Li & Xu, 2019). However, these methods are mainly developed for assessing upper-body postures. By contrast, the work presented in this paper takes a step forward and focuses on the whole-body postural assessment from images, targeting general application scenarios, such as manufacturing and open workplaces.

The proposed method is shown in Fig. 1, which takes as input a single color image and predicts risk scores directly without the explicit estimation of joint locations. That is, rather than assessing the upper-body posture alone as existing work (Li et al., 2020; Mehrizi, Peng, Xu, Zhang, & Li, 2019), six body parts are specifically assessed according to REBA and scores of individual body parts are then aggregated to form a final risk score of the whole-body.

Yet, the whole-body postural assessment from images is challenging due to the possibly large variance of the observed images corresponding to different REBA scores, as illustrated in Figs. 2 and 3; postures with varying REBA scores, on the other hand, may turn out to have similar appearances and vice versa. This necessitates the extraction of discriminative features at the fine-grained level to tackle the variance of the images such as differences in scale and spatial configurations of the body parts.

In this work, an attention-based approach is adopted to extract features from the whole image and as well as salient regions that correspond to body parts. Specifically, a convolutional neural network (CNN) is used to extract global but relatively coarse features. A subsequent attention-module learns to predict multiple attention regions given the coarse features and produce fine-grained features from local regions. By jointly considering global and local fine-grained

* Corresponding author.
E-mail addresses: zewei.ding.kk@gmail.com (Z. Ding), wanqing@uow.edu.au (W. Li), jiey@uow.edu.au (J. Yang), philipo@uow.edu.au (P. Ogunbona), qlhlaa@163.com (L. Qin).

https://doi.org/10.1016/j.eswa.2023.122391
Received 13 February 2023; Received in revised form 27 October 2023; Accepted 27 October 2023
Available online 31 October 2023
0957-4174/© 2023 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

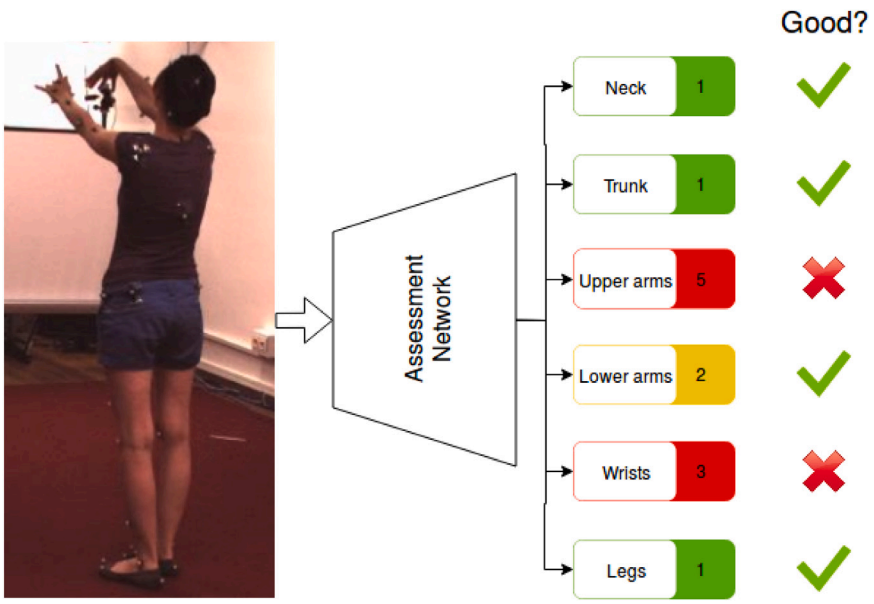


Fig. 1. A person posture is assessed into six REBA scores (each of which corresponding to a specific body part) and scores of individual body parts are later combined to an aggregated risk score of the whole-body.



Fig. 2. Four pairs of samples are illustrated. In each pair, the risks are different while the appearances look similar. From left–right and top-down, the risks of upper arms, trunk, legs, and twists are the same.

information, the proposed network is able to assess the risk of a posture effectively. The attention mechanism adopted in the network is specifically designed so that it not only extracts fine-grained features from key regions, but also enables the model training based on the model prediction loss instead of additional losses like triplet rank loss used previously (Ding et al., 2019).

In order to have an effective posture representation, the spatial relationships among salient regions (or body parts) are encoded as features. In previous methods (Ding et al., 2019), features are extracted using Histogram-Oriented Gradients (HOG) or convolutional operations, where sliding windows are used to obtain local features at each pixel. However, such features fail to encode dependencies among pixels that are at distance. In this work, we propose to explicitly reason dependencies among all pixels (of features), and a novel module, referred to as spatial relation reasoning, is further proposed for this purpose. Experimental results have verified the effectiveness of the proposed methods, including multi-region attention and spatial relation reasoning.

Due to the lack of publicly available datasets for the study of whole-body postural assessment, a new dataset, referred to as pH36M,[1] is created by re-annotating the Human3.6M dataset (Ionescu, Carreira, & Sminchisescu, 2014) which was originally created for the pose estimation. The videos are sampled into frames that are then annotated using the REBA criteria (Hignett & McAtamney, 2000). Specifically, most of the REBA variables are computed from skeleton data accompanying the Human3.6M dataset according to their definition. Variables for

[1] pH36M will be made publicly available after acceptance of the paper.

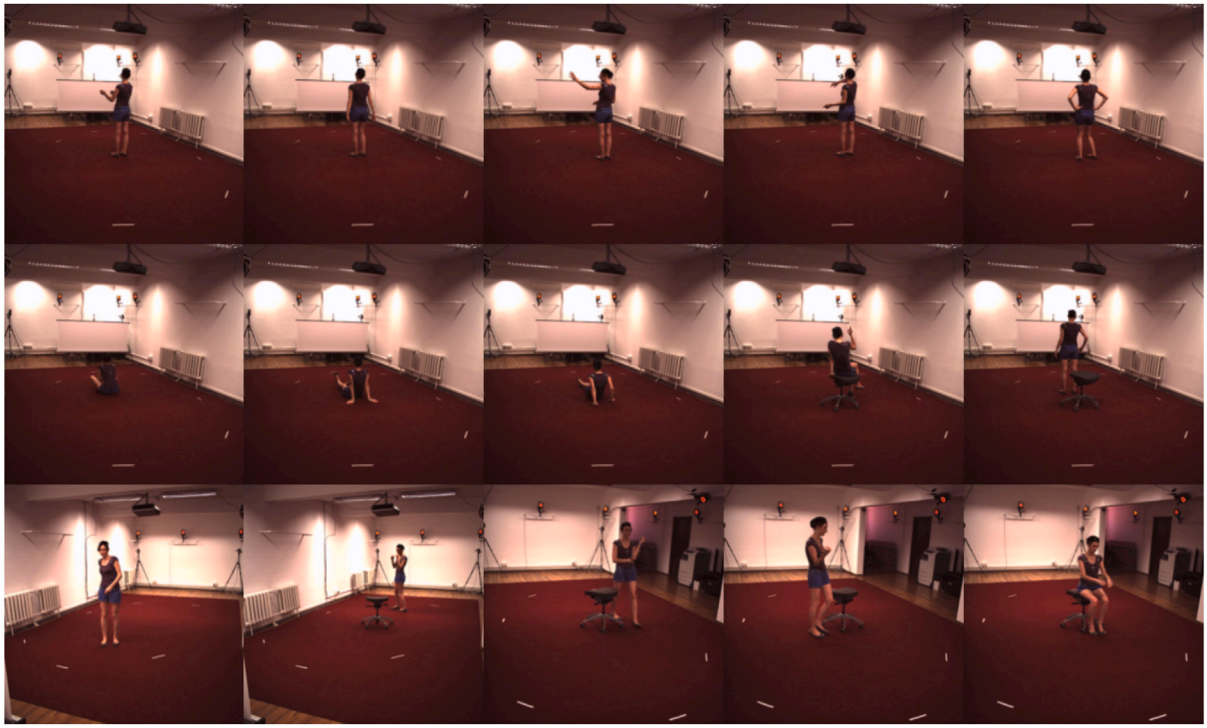


Fig. 3. The postures shown by the 15 images all have a REBA score of 3. The same risk could have various appearances, due to different appearance, body sizes, and viewpoints.

other postures without skeleton data, such as "sitting", are estimated by experienced observers using visual metrics derived from color images (with details provided in Section 4.1).

In summary, the primary contributions of this paper include:

- **Introduction of a Novel Multi-Path CNN:** We propose a unique multi-path CNN for automatic whole-body postural assessment from single images. Each path of the CNN assesses one body part, and all paths are integrated at the feature level to evaluate the entire body. Additionally, we have developed a new loss function to train the entire network end-to-end.
- **Spatial Relation Reasoning (SRR) Module:** A new module, named Spatial Relation Reasoning (SRR), has been introduced. It aims to implicitly understand the spatial relationships between body parts, essential for postural assessment. We also enhanced the Spatial Transformer attention network with a novel loss function. This function constrains the spatial location of attention regions, bolstering the effective assessment of individual body parts.
- **Creation of a Multi-View and Multi-Modalities Dataset:** We have curated a dataset by re-purposing the existing H36M dataset meant for human pose estimation. Our dataset comprises images from different perspectives along with their corresponding 2D and 3D skeletons. To our knowledge, it is the first dataset of its kind tailored for whole-body postural assessment. We intend to make this dataset publicly available upon the paper's acceptance.
- **Experimental Validation:** Our validation process utilized DenseNet-121 and ResNet-50 as the foundational CNNs. We compared various network configurations and assessment using 2D and 3D skeletons. The results affirm the efficacy of our proposed method.

The rest of the paper is organized as follows. Section 2 gives a brief account of background research, including the postural assessment and existing fine-grained feature extraction techniques. Section 3 details the proposed algorithm, with the attention-enhanced convolutional neural network and the spatial relation reasoning. Section 4 discusses the implementation issues (including the pH36M dataset) and evaluates the performance of the proposed algorithm, followed by concluding remarks given in Section 5.

2. Related work

2.1. Postural assessment

In general, methods for postural assessment can be categorized into three approaches, direct measurement based, observation based, and skeleton based, according to how relevant variables (e.g. angles and distances) are measured.

In the direct measurement based approach, instruments including goniometer, inclinometer, potentiometer, and flexicurve are usually employed (Fernandez-de Las-Penas, Alonso-Blanco, Cuadrado, & Pareja, 2006; McVeigh, Murray, Heckman, Rawal, & Peterson, 2016; Youdas, Carey, & Garrett, 1991). When using these tools, one or multiple operators are required to collect and record data. It is inconvenient, inconsistent and high-cost and, thus, infeasible to implement in real workplaces at a large scale. Recently, new techniques have been emerging to automate the measurement of variables. For instance, Motion capture systems (MoCap) (Battini, Persona, & Sgarbossa, 2014; Muller, Pontonnier, Robert-Lachaine, Dumont, & Plamondon, 2020), including optical and non-optical systems, can capture human skeletons from which variables are calculated. Optical systems usually need ample space for setup and are limited in laboratory environment. Non-optical systems such as optoelectronic systems, electromagnetic systems, electromechanical systems, and inertial systems (Mellone, Palmerini, Cappello, & Chiari, 2011; Mundt et al., 2019; Poh, Swenson, & Picard, 2010) have gained attention recently. However, they suffer from time-varying biases and noise, which lead to a quick drift of the captured data after a short time (Filippeschi et al., 2017).

In the observation based approach, subjects' movements are visually examined by trained raters either on location in the field or via recordings (Lowe, Weir, & Andrews, 2014). Some tools have been developed, including such as Ovako Working Postural Assessment System

(OWAS) (Karhu, Kansi, & Kuorinka, 1977), Loading on the Upper-Body Assessment (LUBA) (Kee & Karwowski, 2001), Rapid Upper-Limb Assessment (RULA) (McAtamney & Corlett, 1993), Rapid Entire Body Assessment (REBA) (Hignett & McAtamney, 2000), Occupational Repetitive Actions (OCRA) (Occhipinti & Colombini, 1996), etc. This approach needs professional raters to perform analysis onsite or on the recorded data. Its drawbacks include time consuming and results being subjective because of inter- and intra-rater variance.

Recent years, there is a trend of performing postural assessment directly based on skeletons, either obtained directly from MoCap devices (Diego-Mas & Alcaide-Marzal, 2014; Plantard, Shum, Le Pierres, & Multon, 2017) (e.g. Kinect (Microsoft, 2014)) or estimated from images using pose estimation algorithms (Li et al., 2020; Li & Xu, 2019; Mehrizi et al., 2019, 2018, 2017; Ray & Teizer, 2012). Currently, the state-of-the-art pose estimation methods could achieve reasonable mean per joint position error (MPJPE) (Chen, Tian, & He, 2020) for motion classification, but the estimated poses are not accurate enough for postural assessment, especially when joint angles are close to angle thresholds defined in the criteria, e.g. REBA. The work by Ding et al. (2019) was the first attempt in an automatic upper-body postural assessment using color images rather than skeletons. The advantage of image-based approach over skeleton-based approach has been also justified by experimental results as given in Section 4.

2.2. Attention-based fine-grained feature extraction

Fine-grained features are often extracted based on attention mechanism as inspired by how humans observe images. Attention mechanism has been applied in many computer vision tasks and achieved promising results, including image captioning (Xu et al., 2015), image segmentation (Zhao et al., 2018), image classification (Ghorbel, Ammar, Kessentini, & Jmaiel, 2022; Wang & Girshick, 2018), and fine-grained object recognition (Fu, Zheng, & Mei, 2017; Jaderberg et al., 2015). Basically, attention can be either *hard* or *soft*.

In hard attention, usually only the image region of interest will be fed into the next step for analysis while the rest is filtered out by an attention module or generator. In Xu et al. (2015), a hard-attention approach is proposed for image captioning, which consists of an encoder for feature extraction and a decoder for sentence generation based on features. In the encoder, a sequence of attention regions are sampled following a multinomial distribution, from which a sequence of features are extracted which are mapped to sentences by a decoder. Lam, Mahasseni, and Todorovic (2017) (HSnet) proposed a search-based method for fine-grained object classification. In the method, multiple bounding boxes are searched as attention regions that either correspond to body parts or are visually diverse. These methods use reinforcement learning to train the attention generator based on additional losses apart from classification loss.

Jaderberg et al. (2015) (ST-CNN) introduce spatial transformer networks for fine-grained object classification by extracting features from several image patches. In the network, a spatial transformer (ST) module is used which could automatically locate attention regions and is trained based on only class labels without extra parts annotations. Similarly, the work reported in Fu et al. (2017) repeats the localization processes multiple times to obtain multi-scale attention regions and achieve promising results for fine-grained object recognition. There are mainly two drawbacks in this approach. First, usually rigid shapes are used as attention regions, the accuracy of regions is limited by forms and numbers of shapes which are set empirically. Second, useful information contained in other regions is totally ignored, that is, attention regions are generated and treated independently.

On the other hand, in soft attention, an entire image will be considered in the following analysis but with different importance weights on different regions of the image. Usually, a weight map is generated where pixel values indicate its importance. In some works (Kim, Jun, & Zhang, 2018; Wang et al., 2017), the attention map is generated from same-layer features by down-sampling and up-sampling repeatedly, and applied as a residual to the original branch, so-called residual attention. In the non-local network (Wang & Girshick, 2018), the correlations among pixels are encoded as attention map and multiplied onto same-layer features. The work by Zhao et al. (2018) extends this idea by exploiting correlations of features spatial- and channel-wise. Compared to hard attention, soft attention is able to handle uncertainty. However, unlike hard attention where the parameters of the attention regions (e.g. number and size of regions) are usually explicitly constrained, weight maps are much less constrained. This may lead to difficulties in training and poor generalization.

2.3. Spatial relation reasoning

Reasoning about spatial relation of objects in images has been researched over several years (Lake, Ullman, Tenenbaum, & Gershman, 2017; Logan & Sadler, 1996). Relational representation could be hand-crafted (“up/below”, “left/right”) or learnable like spatial templates, graph model, and embedding representation (Santoro et al., 2017). The representation often relies on the localization of objects, including structured prediction and heatmap regression. Localization-based methods are limited in localization precision and the scale of objects they could handle. This makes them unsuitable for the scenario presented by the current postural assessment since it requires reasoning with relations of small-scale objects, i.e. body parts.

The pixel-wise relational reasoning approach is adopted in this work and is closely related to relational networks (RNs) (Santoro et al., 2017). However, the RN learns relations across multiple explicit objects, and the relations are explicitly provided and relatively simple. In postural assessment, the relations between body parts are to be reasoned, which is more challenging due to diverse body parts and complex relations that contribute to the assessment score. In addition, the relation labels are not provided, the network needs to learn relations implicitly associated with REBA scores.

3. Proposed method

The proposed network is shown in Fig. 4. It comprises of a front-end feature extractor and a back-end ordinal regressor(s). The feature extractor consists of a **base** branch, an **attention module**, and a set of **spatial relation reasoning (SRR)** branches. The base branch can be any conventional convolutional network, and ResNet50 is adopted here as an example. The attention module predicts attention regions based on high-level features of the base branch which provides global information. Given the predicted attention regions, the corresponding regions in low-level feature maps are sampled and fed into SRR branches. Finally, the output features of base and SRR branches are concatenated to form the final posture descriptor.

During training, the features extracted by the base and SRR branches and final concatenated features are fed into different independent ordinal regressors. Details refer to Section 3.4. In testing, the output of the ordinal regressor connected to the final feature is used. In the following, the attention module, SRR branch, and ordinal regressor are described in detail.

3.1. Predicting attention regions with spatial transformers

In subjective assessment from recorded videos, ergonomists usually zoom into specific body parts in order to deal with large among subjects, the key idea is to pay attention to distinguishable details.This is mimicked by the spatial transformer (ST) (Jaderberg et al., 2015) to help the network zoom at multiple discriminative regions in the proposed method. A spatial transformer consists of a localization network, a grid generator, and a sampler, which are detailed in the following.

The localization network predicts the locations of attention regions from input images or features. In particular, given the feature map

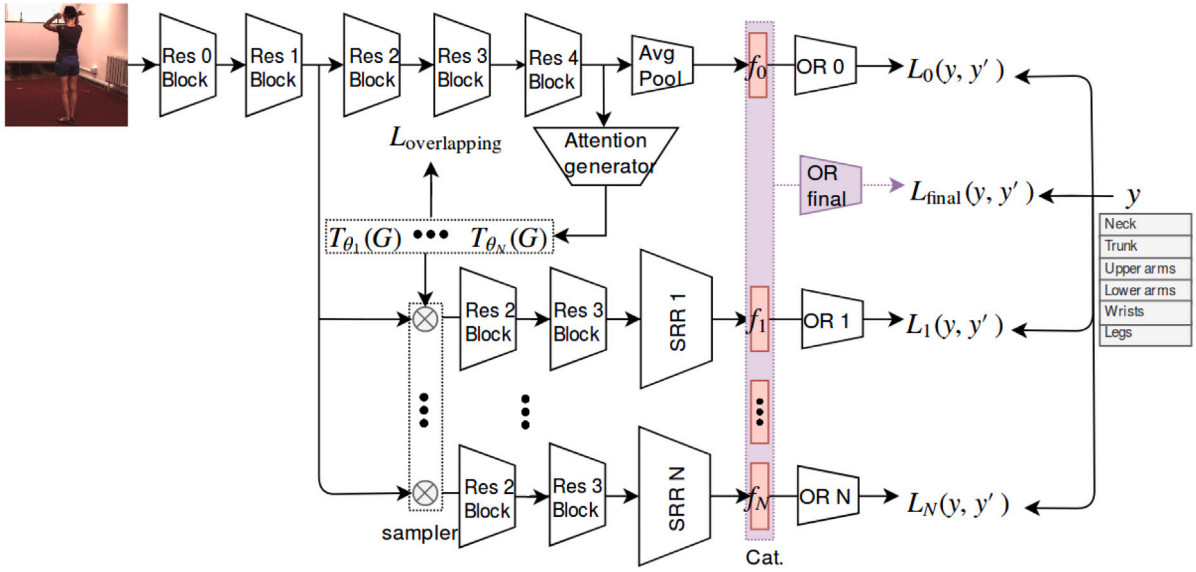


Fig. 4. Architecture of the proposed network. Each ResBlock represents a set of convolutional layers with residual connections. The $T_{\theta_i}(G)$ denotes the sampling grid generated based on θ_i.

$U \in R^{W\times H\times C}$, where W, H, C are the width, height, and number of channels, the localization net f_{loc} predicts N transformation matrices

$$f_{loc}(U) = \{\Theta_1, \Theta_2, \ldots, \Theta_N\} \tag{3.1}$$

where N is the number of attention regions, and Θ_n is a matrix parameterized by θ. An affine transformation is used here since the goal is to find attention regions. In addition, REBA scores are mainly determined by spatial relations among body joints (e.g. angles, relative distances). Hence in order to preserve spatial relations, only cropping has been used in this work since other transformations may cause geometric distortions and, hence, change the relations of joints. Moreover, the aspect ratio is kept the same during the transformation. The transformation matrix is given as

$$\Theta_n = \begin{bmatrix} \theta_1^n & 0 & \theta_2^n \\ 0 & \theta_1^n & \theta_3^n \end{bmatrix}, \forall n \in N \tag{3.2}$$

where θ_1^n determines the region size and θ_2^n, θ_3^n determine its horizontal and vertical positions.

Given the transformation matrices, a sampling grid could be generated $T_{\theta_n}(G)$, where $G = \{G_i\} = \{(x_i, y_i)\}$ and i denotes the ith pixel. The sampling grid defines the relations between pixels of target and source feature maps. Specifically,

$$\begin{pmatrix} x_i^s \\ y_i^s \end{pmatrix} = T_{\theta_n}(G_i) = \begin{bmatrix} \theta_1^n & 0 & \theta_2^n \\ 0 & \theta_1^n & \theta_3^n \end{bmatrix} \begin{pmatrix} x_i^{tn} \\ y_i^{tn} \\ 1 \end{pmatrix}, \tag{3.3}$$

$$\forall n \in N, \forall i \in \{1, 2, \ldots, HW\}$$

where (x_i^s, y_i^s) is the location of ith pixel in source feature map, (x_i^{tn}, y_i^{tn}) is the location of ith pixel in nth target feature map, N is the number of attention regions.

The sampler applies to obtain a set of sampling points given in $T_\theta(G)$, on source feature map $U \in R^{H\times W\times C}$ and generates target image $V \in R^{H'\times W'\times C}$. Note that the channels are identically sampled to preserve spatial consistency in channels. For each point in target feature map V_i^c, the sampler get values from specific locations $\{(x_i^s, y_i^s)\}$ in source feature map U. In this work, bilinear interpolation is used, as shown in Eq. (3.4).

$$V_i^c = \sum_m^H \sum_n^W U_{m,n}^c \max(0, 1 - |x_i^s - n|) \max(0, 1 - |y_i^s - m|), \tag{3.4}$$

$$\forall i \in \{1, 2, \ldots, H'W'\}, \forall c \in \{1, 2, \ldots, C\}$$

The sampling kernel has partial derivatives defined so it could learn to predict attention regions (Eqs. (3.6) and (3.7)) as well as learn to extract features from regions (Eq. (3.5)). It is easy to calculate $\frac{\partial x_i^s}{\partial \theta}$, and parameters of f_{loc} could be updated.

$$\frac{\partial V_i^c}{\partial U_{m,n}^c} = \sum_m^H \sum_n^W \max(0, 1 - |x_i^s - m|) \max(0, 1 - |y_i^s - n|) \tag{3.5}$$

$$\frac{\partial V_i^c}{\partial x_i^s} = \sum_m^H \sum_n^W \max(0, 1 - |y_i^s - n|) \begin{cases} 1 & \text{if } n \geq x_i^s \\ 0 & \text{if } |n - x_i^s| \geq 1 \\ -1 & \text{if } n < x_i^s \end{cases} \tag{3.6}$$

$$\frac{\partial V_i^c}{\partial y_i^s} = \sum_m^H \sum_n^W \max(0, 1 - |x_i^s - n|) \begin{cases} 1 & \text{if } m \geq y_i^s \\ 0 & \text{if } |m - y_i^s| \geq 1 \\ -1 & \text{if } m < y_i^s \end{cases} \tag{3.7}$$

In Jaderberg et al. (2015), a color image is fed as input to localization network, and f_{loc} is implemented as a conventional CNN such as the Inception module with batch normalization. There are mainly two issues. First, the localization network has numerous parameters that are updated based on Eqs. (3.6) and (3.7). It has been shown that to train the whole network well it needs a great amount of data otherwise there is a possible over-fitting problem (Jaderberg et al., 2015). Second, features are extracted from only attention regions and information in other regions is ignored, decreasing the effectiveness of extracted features. Later we shall show in experiments that such architecture works poorly on postural assessment problems.

In particular, as shown in Fig. 4, the output of ResBlock4 in branch 0 is fed into the localization network for generating the attention regions. Localization network f_{loc} is implemented by a 1×1 convolutional layer for channel reduction followed by a multi-layer perceptron (MLP). There is no pooling layer in the localization network in order to preserve spatial information. In this way, the localization network has much fewer parameters to be updated, and are less prone to the over-fitting problem. In addition, instead of applying hard attention mechanisms on input images (Fu et al., 2017; Jaderberg et al., 2015), we propose to apply attention regions on intermediate features maps. It has been shown that selecting hard regions on higher-level features is equivalent to having soft attentions on lower-level features (Guo, Wang, Chen, & Zhang, 2017; Sun, Zheng, Yang, Tian, & Wang, 2018).

To enhance discriminative power, the network should extract features from different regions. Several metrics have been proposed to

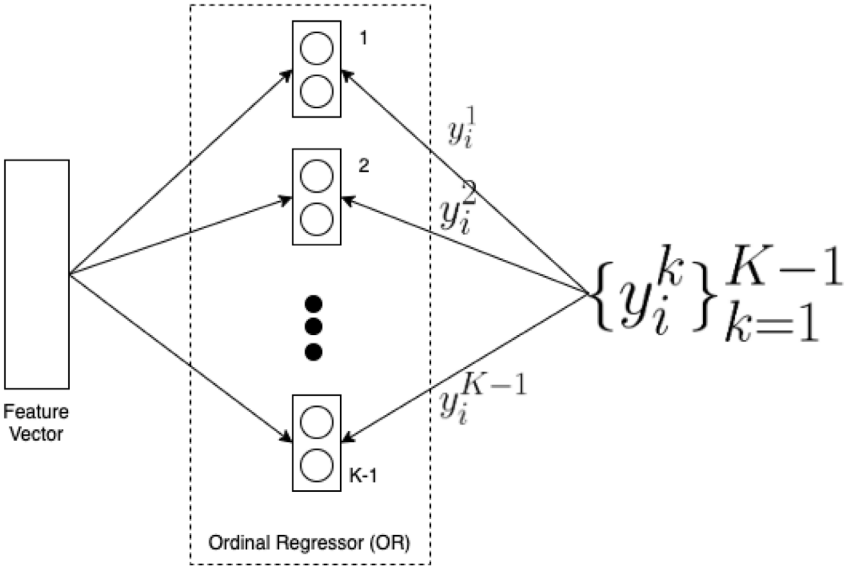


Fig. 5. The ordinal regressor used in the proposed network. Each unit in the ordinal regressor module represents a two-neuron fully-connected layer.

measure region diversity, including determinantal point processes (DPP) (Lam et al., 2017), cross-entropy of feature (Liu & Ferrari, 2017), etc. For simplicity, overlapping among regions is penalized as regularization in this work.

$$\begin{aligned} L_{OL} &= \sum_{i=1}^{N-1}\sum_{j=i+1}^{N} R_i \bigcap R_j \\ &= \sum_{i=1}^{N-1}\sum_{j=i+1}^{N} [\min(x_2^i, x_2^j) - \max(x_1^i, x_1^j)] \\ &\times [\min(y_2^i, y_2^j) - \max(y_1^i, y_1^j)] \\ &= \sum_{i=1}^{N-1}\sum_{j=i+1}^{N} [\min(\theta_2^i + \theta_1^i, \theta_2^j + \theta_1^j) - \max(\theta_2^i - \theta_1^i, \theta_2^j - \theta_1^j)] \\ &\times [\min(\theta_3^i + \theta_1^i, \theta_3^j + \theta_1^j) - \max(\theta_3^i - \theta_1^i, \theta_3^j - \theta_1^j)] \end{aligned} \tag{3.8}$$

where N denotes the number of regions; R_i, R_j denote the attention regions calculated from θ, $\bigcap$ is region overlap operation. The box region R_i could be represented by two points, top-left point (x_1^i, y_1^i) and bottom-right point (x_2^i, y_2^i). This loss is directly applied to transformation matrix parameters Θ, and the partial gradients $\frac{\partial L_{OL}}{\partial \theta}$ is easy to calculate. Hence, during the training, the weights of the localization network are updated not only based on errors back-propagated from prediction loss (ordinal regression loss as described below), but also based on overlapping loss. The former guides the network to find regions that contain discriminative features, and the later enforces the network to search for diverse regions.

3.2. Spatial relation reasoning

To determine the REBA scores, the spatial relations of body parts or joints are required. Since there are no explicit body parts/joints detected, the relations between each pixel pairs in feature maps are used instead for reasoning. Each pixel of the feature map is treated as a potential joint and used in reasoning about a dense spatial relation. Specifically, the relations of each pair of pixels of the feature maps are encoded into feature representations f, described as

$$f = \sum g(x_i, x_j), \forall x_i, x_j \in X' \tag{3.9}$$

where $g(,)$ is a mapping function, x_i, x_j are pixels of feature map X'. In this work, X' is extracted by ResBlock 2 & 3 from target feature maps X^t, that is the output of the ST module. The mapping function $g(x, y)$ is implemented with a 1×1 convolution kernel whose input is the concatenation of x and y. Other mapping functions such as Gaussian kernel may also be used.

3.3. Ordinal regression

As mentioned above, REBA scores are discrete and ordered. For example, trunk has five scores denoting different risk levels with $r_5 \succ r_4 \succ \cdots \succ r_1$ where the symbol $\succ$ denotes risk ordering. Although nominal classification (Ding, Wang, Ogunbona, & Li, 2017; He, Zhang, Ren, & Sun, 2016; Huang, Liu, Van Der Maaten, & Weinberger, 2017) and metric regression (Agarwal & Triggs, 2004; Luvizon, Picard, & Tabia, 2018) methods could be used to predict REBA scores, they have limitations in predicting ordinal categories. For nominal classification, ordinal information is not exploited and performance is limited. For metric regression, training models with several discrete numbers are prone to over-fitting. In addition, discretizing continuous outputs may introduce extra errors.

Thus, in this work, the prediction is formulated as an ordinal regression to exploit ordinal cues contained in risks and yield better performance. Similarly to Niu, Zhou, Wang, Gao, and Hua (2016), we take the task decomposition approach (Li & Lin, 2007) for ordinal regression, where the whole task is decomposed into multiple simpler tasks (e.g. binary classification). In Section 4, we show that ordinal regression has superior performance than nominal classification or metric regression.

Given a training set $D = \{x_i, y_i\}_{i=1}^N$, where $x \in X$ and $y \in Y = \{r_1, r_2, \ldots, r_K\}$ with orders $r_K \succ r_{K-1} \succ \cdots \succ r_1$, we first decompose each target y_i into $K-1$ binary targets $\{y_i^k\}_{k=1}^{K-1}$,

$$y_i^k = \begin{cases} 1 & \text{if } (y_i > r_k) \\ 0 & \text{otherwise} \end{cases} \tag{3.10}$$

For each pair $\{x_i, y_i^k\}$, a binary classifier f_k is trained. Hence, the original ordinal regression is decomposed into $K-1$ binary classification sub-problems which are simultaneously solved. In this work, each binary classifier is implemented as a single fully connected layer with two neurons, as shown in Fig. 5. The $K-1$ binary classifiers share the same feature representation and thus the correlation of different tasks could be exploited, which benefits the final performance.

Each binary classifier is trained with cross-entropy loss, and the final loss of an ordinal regressor is given as,

$$L_{OR}(y, x; K) = \sum_{k=1}^{K-1} -y^k \log f_k(x) - (1 - y^k) \log(1 - f_k(x)) \tag{3.11}$$

where x is the feature vector, f_k is the kth binary classifier, $\{y_i^k\}_{k=1}^{K-1}$ is the decomposed binary labels, K is the number of levels.

During testing, the REBA score is estimated as

$$y' = 1 + \sum_{k=1}^{K-1} argmax_i(f_k(x)) \tag{3.12}$$

where $argmax_i()$ refers to the function of getting index i with maximum value, and $i \in \{0, 1\}$.

3.4. Training the network

In the network as shown in Fig. 1, given N attention regions, there are $N + 2$ branches, including a base branch (branch 0) for extracting global features, N branches (branch $1 - N$) encoding local information, and a final branch (branch final) predicting risks based on features of all branches. In training the network, each branch is trained with a specific loss.

$$L_B(y, x) = \sum_{b \in \{parts\}} L_{OR}(y, x; K_b) \tag{3.13}$$

where x is feature of a branch, $\{parts\}$ refers to six body parts {neck, trunk, upper arms, lower arms, wrists, legs}, K_b denotes the risk score range of body part b, L_{OR} is the ordinal loss in Eq. (3.11). Being trained with losses of six body parts, each branch is trained to extract features that are useful for tasks. Together with overlapping loss, the network is trained to extract features from salient and diverse regions.

The final loss for end-to-end training of the whole network can be written as,

$$L = L_{final} + \alpha L_0 + \beta \sum_{B=1}^{N} L_n + \gamma L_{OL} \tag{3.14}$$

where L_{final}, L_0, and $\{L_n\}_{B=1}^{N}$ are losses of different branches (Eq. (3.13)), and L_{OL} is the overlapping loss (Eq. (3.8)) applied on localization network as a regularization; α, β, γ are weight items.

4. Experiments

4.1. pH36M dataset

Since there are no datasets publicly available for the study of whole-body postural assessment, a new dataset called pH36M is created in this work by re-annotating the Human3.6M dataset (Ionescu et al., 2014), a widely used dataset for pose estimation. In the Human3.6M dataset, subjects are recorded by 15 sensors, including four high-resolution digital cameras, one time-of-flight camera, and an optical MoCap system (10 motion cameras) that are synchronized. Therefore, for each recording, the system outputs image data, pose data, range data, and scanner data. Among them, image data refers to color images from four cameras that are set at four corners of the collection space. Pose data includes a 24-joint 3D skeleton provided by the MoCap system and four 2D skeletons corresponding to four images which are the mappings of the 3D skeleton on the camera planes.

In the Human3.6M dataset, there are 17 scenarios including discussion, smoking, taking photos, talking on the phone, and so on. They are performed by eleven professional actors (6 male, 5 female). For each scenario, each actor performs twice, resulting in a total of $17 * 2 * 11 * 4 = 1496$ recordings. With horizontal flipping as data augmentation, there are a total of 3.6 million frames in the dataset, so-called Human3.6M.

We re-annotate the Human3.6M dataset for the study of postural assessment. The new dataset consists of images from the Human3.6M dataset and corresponding labels of postural risks calculated based on pose data provided in Human3.6M. We name the new dataset as pH36M for simplicity.

The re-annotation is in a frame-based manner. For risk calculation, "Rapid Entire Body Assessment" (REBA (Hignett & McAtamney, 2000)) is adopted. The variables to be measured and measurement methods are given in Table 1. The procedure is shown in the following. A GUI-based tool is developed using Matlab to assist the annotation.

Table 1
REBA variables used in annotation. All angles are calculated from 3D skeleton data, – those auxiliary variables do not belong to specific body parts.

Groups	Body parts	Variables	Measurement methods
Group A	Head	Flexion/Extension	Angles
		Twisting/Side flexed	Angles
	Trunk	Flexion/Extension	Angles
		Twisting/Side flexed	Angles
	Legs	Bilateral weight bearing	Angles
		Flexion	Angles
		Sitting	Visual Examination
	–	Load	Visual Examination
	–	Force	Visual Examination
Group B	Upper Arms	Flexion/Extension	Angles
		Arm abducted/rotated	Angles
		Should raised	Angles
		Supported	Visual examination
	Lower Arms	Flexion	Angels
	Twists	Flexion/Extension	Angles
		Deviated/Twisted	Angles
	–	Coupling	Visual Examination
Group C	–	Static	Visual Examination
		Repeated	Visual Examination
		Unstable	Visual Examination

1. Select an instance from the Human3.6M dataset. An instance contains a 3D skeleton model and four color images sampled from video clips corresponding to four cameras.
2. Obtain the score of Group A.
   (a) Obtain scores of individual parts, including head (neck), trunk, and legs. Variables are measured by either calculating the angles and distances based on the 3D skeleton or visually examining the color images. Details of measuring methods are given in Table 1.
   (b) Obtain the joint score using the lookup table A in the original paper (Hignett & McAtamney, 2000).
   (c) Measure auxiliary variables including Load and Force and obtain scores.
   (d) Obtain group score by adding the joint score and auxiliary scores.
3. Obtain the score of Group B.
   (a) Obtain individual scores of each body parts, including Upper Arms, Lower Arms, and Twists.
   (b) Obtain the joint score using lookup table B.
   (c) Measure score of coupling.
   (d) Obtain group score by adding the joint score and coupling score.
4. Obtain the joint score of Group A & Group B using the lookup table C.
5. Obtain auxiliary scores, including static, repeated, and unstable.
6. Obtain the final risk score of the entire body by adding auxiliary scores and joint score.

The official Human3.6M has two parts for training and test. In this work, the publicly available training set is re-annotated, which contains 16 scenarios and 7 subjects (indexes of 1, 5, 6, 7, 8, 9, 11). For each

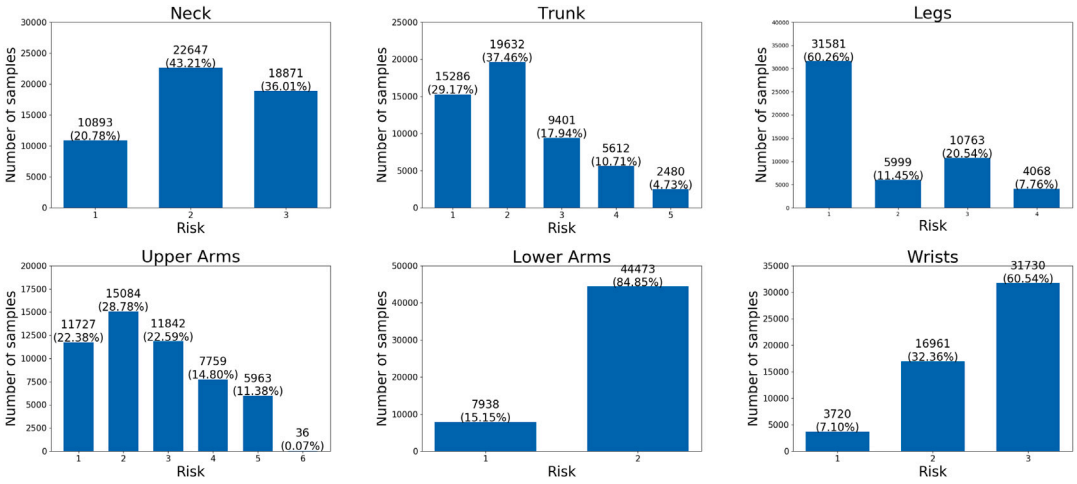


Fig. 6. The distribution of six body part scores. The number above each bar denotes the number of samples of the corresponding risk.

recording, one out of ten frames is annotated. In total, there are 52411 instances sampled and annotated. The risk distributions of six body parts are shown in Fig. 6.

4.2. Setup

Images are cropped into square boxes which enclose subjects only. The square box cropping aims at preserving the aspect ratio of the subjects, thus preserving the spatial relationships among joints. Cropped patches are resized to a size of 256 × 256 pixels as commonly used in image classification. Increasing the resolution could improve algorithm performance, but we opted this format because the aim is to compare performance with baseline methods. Data are then augmented by random horizontal flipping. Since subjects cover most of the images, there is no further cropping in augmentation here to preserve all necessary information.

In addition, the dataset is divided into three groups according to camera viewpoints. The viewpoint is defined as the angle between the orientation of the hip joint and the direction from the hip joint to the camera. There are three categories, "front-view" ($\theta \leq 45°$), "side-view" ($45° < \theta < 135°$), and "back-view" ($135° \leq \theta$). In total there are 80358 front-view images, 49516 side-view images, and 79770 back-view images. Each group is divided into two subsets with ratio 80/20 for training and test.

The proposed network and the chosen baselines are implemented in PyTorch 0.4.0 and optimized using mini-batch SGD. The feature extractors are initialized with the weights pre-trained on ImageNet. The network is trained for 40 epochs with the base learning rate initially set to 0.01 and multiplied by 0.1 at epochs (11, 21, 31, 36). The base learning refers to feature extractors, while the learning rate for ST fc_{loc} is 1e-4 of the base learning rate, which is similar to those in Jaderberg et al. (2015). In experiments, the weight items are set as $\alpha = 1, \beta = 1, \gamma = 0.001$ respectively.

4.3. Baselines

Besides the proposed method, we also evaluate the following methods as baselines. For all the baselines, the last fully-connected layer is replaced with ordinal regressors.

- MLP-Skel2D. An MLP is trained and tested on synchronized 2D skeleton data. The MLP has 3 hidden layers with each having 2048 neurons. This baseline could be considered as the upper bound of methods based on 2D skeletons estimated by pose estimation methods such as OpenPose (Cao, Simon, Wei, & Sheikh, 2017).
- MLP-Skel3D. An MLP is trained and tested on synchronized 3D skeleton data. The MLP has 3 hidden layers with each having 2048 neurons. This baseline could be considered as the upper bound of the skeleton-based approach.
- RN50. The conventional ResNet50 (He et al., 2016).
- DN121. The conventional DenseNet121 (Huang et al., 2017).

4.4. Performance metrics

Two performance metrics are used in this work. The mean absolute error (MAE) and Cohen' Kappa coefficient (κ) (Cohen, 1960). Accuracy has not been used as a performance metric because it is not reliable for unbalanced risk distribution (Fig. 6). For example, the lower arm has 2 risk levels, and 84.5% of the samples have a risk of 2. Given a classifier that identifies all samples into risk 2, it has accuracy 84.5% which is meaningless. Instead, Cohen's Kappa coefficient could handle the unbalanced distribution better by considering the underlying distribution in evaluation. In the above case, the classifier has a $\kappa = 0$ which means the predictions are not correlated with ground-truth targets at all.

- Mean absolute error (MAE).

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|y_i^* - y_i| \tag{4.1}$$

  where N is the number of test samples; y^* and y are prediction and target respectively.
- Cohen's Kappa coefficient (κ). It measures the agreement between predictions and targets. It accounts for the data distributions and can reflect the differences even if the accuracy is the same. Note that $\kappa \in [-1, 1]$ and a higher number indicates stronger agreement. The coefficient is defined as

$$\kappa = \frac{p_o - p_e}{1 - p_e} \tag{4.2}$$

where p_o is the observation accuracy calculated as conventional accuracy; p_e is the hypothetical probability of chance agreement calculated based on confusion matrix; $p_e = \frac{1}{N^2}\sum_k n_{kp}n_{kt}$, where n_{kp} and n_{kt} are the numbers of prediction and target samples in class k respectively, and N is the total number of samples.

4.5. Results

Baselines are evaluated on front-view data and results are shown in Table 2. The performance of six body parts is given specifically. In addition, for comparison, an overall performance of all six body parts is given as well (last column in the table). It could be seen

Table 2
Results on pH3.6M (front-view). In "Ours-x", x refers to the number of branches.

MAE/κ	Legs	Lower arms	Neck	Trunk	Upper arms	Wrists	Mean
MLP-Skel2D	.157/.740	.056/.791	.291/.560	.238/.676	.169/.790	.149/.713	.177/.712
MLP-Skel3D	.079/.860	.041/.850	.174/.733	.148/.799	.128/.839	.129/.753	.117/.806
RN50	.129/.783	.072/.719	.268/.594	.250/.665	.201/.757	.175/.662	.183/.697
DN121	.123/.797	.059/.782	.226/.661	.221/.701	.167/.796	.161/.695	.160/.739
Ours-1	.121/.803	.060/.775	.239/.639	.234/.688	.165/.797	.149/.712	.161/.736
Ours-2	.119/.804	.057/.782	.241/.639	.236/.683	.160/.802	.150/.712	.161/.737
Ours-4	.112/.816	.056/.788	.240/.643	.234/.686	.150/.815	.155/.702	.158/.742
Ours-6	.112/.815	.056/.788	.245/.633	.234/.687	.152/.813	.153/.704	.159/.740

Table 3
The performance (MAE/κ) of using different attention modules (front-view).

Methods	Legs	Lower arms	Neck	Trunk	Upper arms	Wrists	Mean
RN50	.129/.783	.072/.719	.268/.594	.250/.665	.201/.757	.175/.662	.183/.697
RN50-NL	.129/.785	.063/.753	.245/.628	.227/.694	.175/.786	.158/.690	.166/.723
RN50-ST-6	.409/.427	.147/.035	.434/.318	.579/.205	.863/.098	.304/.433	.456/.253
RN50-ST2–6	.122/.802	.063/.764	.246/.629	.220/.705	.179/.781	.172/.670	.167/.725
Ours-4	.112/.816	.056/.788	.240/.643	.234/.686	.150/.815	.155/.702	.158/.742

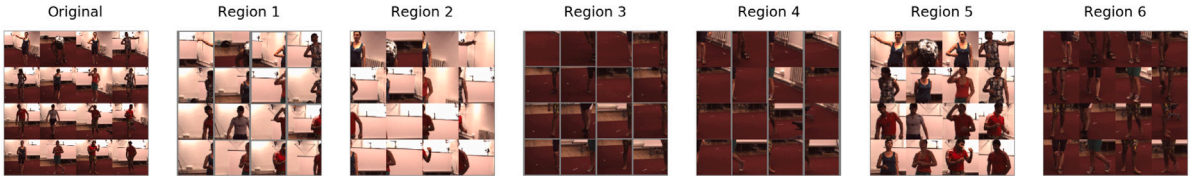


Fig. 7. The input images and corresponding six attention regions predicted by the network. Note that the attention regions are applied on feature maps.

that, of all image-based methods, including RN50, DN121, and the proposed method, the proposed method achieves the best performance for all body parts, with more than 5 percentage point improvement on average. It even achieves better performance for upper arms than 3D skeleton-based MLP-Skel3D.

Several observations can be made. First, all methods achieve better performances than MLP-Skel2D for almost all body parts. It shows that images contain additional useful information than joint positions alone, and assessing posture directly from images has advantages over approaches based on 2D pose estimation methods. Second, the performance of the proposed method increases with the number of attention regions and saturates after four. One more interesting finding is that jointly training six body parts outperforms independent training of each body part. This may be because body parts relationships are learned in joint training, and cues from large body parts such as lower arms could help to extract features of small-scale parts like wrists. More ablation studies are provided in the following.

4.5.1. Contribution of attention-based feature extraction

To verify that the STN module helps in extracting fine-grained features, the proposed method is compared with three methods as described below.

- RN50-NL. The conventional ResNet50 with a non-local module inserted after ResBlock4. Non-local module proves to be able to extract better features by considering the relationships of pixels as attention weights.
- RN50-ST-6. Following the architecture as used in Jaderberg et al. (2015), the input image is fed directly into the localization network for attention generation. The sampler samples six attention regions from input images, and are fed into six independent feature extractors. The features of six branches are concatenated as the final feature for inference. The localization network and six feature extractors are all implemented with ResNet50.
- RN50-ST2-6. Following the architecture as proposed in this work. The network is different from the proposed network in that, there is no loss to the base branch and the parameters of ResBlock3 &4 in the base branch as updated only based on errors back-propagated from the attention regions.

Table 3 shows the results. It can be seen that methods with attention module outperform the conventional ResNet50, except for RN50-ST-6. On one hand, the attention mechanism contributes towards extracting more discriminative features for postural assessment. On the other hand, the poor performance of RN50-ST-6 shows that localization networks with a large number of parameters are hard to train and prone to over-fitting, which is consistent with analysis. By reducing the parameters in the localization network (RN50-ST2-6), the localization network could successfully learn the attention regions and the whole network works. In addition, the superior performance and fewer parameters of the proposed method (Ours-4) compared with RN50-ST2-6 shows that predicting attention regions from high-level features in the base branch is effective.

Examples of attention regions are illustrated in Fig. 7. It can be seen that the network has successfully learned six different regions. The regions do not exactly correspond to six body parts but cover multiple key parts, including lower body, upper body, shoulder. It is reasonable since REBA scores are calculated from relative positions/orientations between other body parts. For example, if the subjects have the same neck orientation but have different trunk orientations, then the neck postures are different and so are neck scores. Hence, to reason neck angles it is necessary to have attention regions covering both trunk and neck.

4.5.2. Contribution of spatial relation reasoning

The performance of networks with and without SRR modules are compared, as described below, and the results are given in Table 4. It could be seen that RN50-SRR achieves better results on average. This verifies the effectiveness of SRR which explicitly reasons about relations among feature pixels.

- RN50. Conventional ResNet50.

Table 4
The performance (MAE/κ) of baseline ResNet50 and that with SRR module (front-view).

Methods	Legs	Lower arms	Neck	Trunk	Upper arms	Wrists	Mean
RN50	.133/.784	.063/.760	.245/.634	.234/.687	.182/.780	.160/.695	.170/.723
RN50-SRR	.125/.795	.064/.757	.244/.635	.227/.688	.180/.782	.163/.690	.166/.725

Table 5
Performances (MAE/κ) of different inferring approaches (front-view).

Methods	Legs	Lower arms	Neck	Trunk	Upper arms	Wrists	Mean
MLP-Skel2D-CLA	.209/.675	.077/.695	.374/.478	.302/.597	.223/.732	.172/.670	.226/.641
MLP-Skel2D-MR	.224/.624	.073/.729	.464/.233	.383/.472	.366/.545	.209/.604	.287/.535
MLP-Skel2D-OR	.157/.740	.056/.791	.291/.560	.238/.676	.169/.790	.149/.713	.177/.712
RN50-CLA	.135/.785	.065/.755	.250/.634	.243/.679	.189/.777	.169/.681	.175/719
RN50-MR	.238/.609	.091/.642	.424/.320	.429/.416	.435/.473	.262/.477	.313/.490
RN50-OR	.133/.784	.063/.760	.245/.634	.234/.687	.182/.780	.160/.695	.170/.723

Table 6
Results on different views in pH36M. The numbers are in the format of MAE/κ. [1]Note that the results for front view are duplicated from Table 2.

Views	Methods	Legs	Lower arms	Neck	Trunk	Upper arms	Wrists	Mean
Side	MLP-Skel2D	.133/.775	.060/.777	.286/.571	.239/.674	.174/.785	.136/.737	.171/.720
	MLP-SKel3D	.081/.859	.038/.861	.179/.725	.149/.797	.127/.841	.130/.753	.117/.806
	RN50	.230/.702	.069/.666	.327/.526	.308/.583	.243/.710	.220/.614	.233/.634
	DN121	.213/.719	.062/.703	.287/.575	.279/.618	.211/.745	.199/.648	.209/.668
	Ours-2	.197/.739	.058/.708	.294/.563	.274/.625	.201/.754	.190/.660	.202/.675
	Ours-4	.196/.741	.058/.713	.311/.547	.272/.627	.196/.759	.185/.671	.203/.676
Back	MLP-Skel2D	.137/.767	.061/.776	.287/.567	.239/.676	.179/.779	.133/.742	.173/.718
	MLP-SKel3D	.076/.866	.040/.853	.180/.724	.153/.792	.126/.842	.129/.754	.117/.805
	RN50	.138/.777	.078/.704	.260/.613	.232/.692	.190/.772	.183/.650	.180/.701
	DN121	.125/.795	.069/.739	.224/.664	.207/.720	.167/.796	.163/.688	.159/.734
	Ours-2	.118/.809	.065/.747	.248/.629	.208/.718	.154/.810	.174/.668	.161/.730
	Ours-4	.115/.813	.064/.749	.245/.633	.206/.720	.152/.812	.166/.681	.158/.735
Front[1]	Ours-4	.112/.816	.056/.788	.240/.643	.234/.686	.150/.815	.155/.702	.158/.742

- RN50-SRR. The conventional ResNet50 with an SRR inserted after ResBlock4. Compared with RN50, it explicitly reasons the spatial relationships of pixels of features out of ResBlock4.

4.5.3. Contribution of ordinal regression

In order to show the effectiveness of the ordinal regressor (OR), it is compared with both nominal classification (CLA) and metric regression (MR). From the results shown in Table 5, ordinal regression has superior performances over metric regression and nominal classification. It shows that leveraging ordinal information contained in the risk scores improves performance.

- RN50-CLA. The task is formulated as a nominal classification problem (He et al., 2016; Huang et al., 2017), with each risk score being a specific class. The network is trained with cross-entropy loss.
- RN50-MR. The task is formulated as a metric regression problem. A fully-connected layer is used to take the feature of ResBlock4 as input and outputs a continuous number of $r \in (-\infty, \infty)$. The network is optimized with mean squared loss. In inference, the $(-\infty, \infty)$ is first divided into K zones, $(-\infty, 1.5), [1.5, 2.5), \ldots, [K - 0.5, \infty)$, and the risks are inferred according to which zone r belongs to.

4.5.4. Performance on different viewpoints

The proposed method is evaluated on different viewpoints, including front-view, side-view, and back-view. Results are given in Table 6. Generally, similar patterns as aforementioned are observed on the other two viewpoints. Of all three viewpoints, methods achieve the worst performance on side-view, and achieve comparable performances on front-view and back-view data.

5. Conclusion

An attention-based convolutional neural network is proposed for postural assessment. The network takes single color images and predicts REBA scores of six body parts. The network consists of a base branch for global feature extraction, a spatial transformer module for multiple attention region prediction, and a corresponding number of branches for fine-grained local feature extraction. Evaluation of the method on the pH36M dataset outperforms the baselines, showing the effectiveness of the proposed network. In addition, preliminary results have shown that postural assessment may provide additional information for analysis of daily activities. However, the network size and the complexity of training and testing increase with the number of branches. In future work, the reduction of complexity of the network will be studied.

CRediT authorship contribution statement

Zewei Ding: Methodology, Software, Validation, Investigation, Writing – original draft. **Wanqing Li:** Conceptualization, Methodology, Formal analysis, Writing – review & editing, Supervision. **Jie Yang:** Methodology, Formal analysis, Investigation, Writing – review & editing. **Philip Ogunbona:** Conceptualization, Writing – review & editing, Supervision. **Ling Qin:** Conceptualization, Methodology, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

Zewei Ding was supported by the International Postgraduate Tuition Award, University of Wollongong, Australia, and CSC (Chinese Scholarship Council) Scholarship.

References

Agarwal, A., & Triggs, B. (2004). 3D human pose from silhouettes by relevance vector regression. In *Proceedings of the IEEE conference on computer vision and pattern recognition, vol. 2* (pp. II–882). IEEE.

Battini, D., Persona, A., & Sgarbossa, F. (2014). Innovative real-time system to integrate ergonomic evaluations into warehouse design and management. *Computers & Industrial Engineering*, *77*, 1–10.

Cao, Z., Simon, T., Wei, S.-E., & Sheikh, Y. (2017). Realtime multi-person 2D pose estimation using part affinity fields. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 7291–7299).

Chen, Y., Tian, Y., & He, M. (2020). Monocular human pose estimation: A survey of deep learning-based methods. *Computer Vision and Image Understanding*, *192*, Article 102897.

Cohen, J. (1960). A coefficient of agreement for nominal scales. *Educational and Psychological Measurement*, *20*(1), 37–46.

Diego-Mas, J. A., & Alcaide-Marzal, J. (2014). Using Kinect™ sensor in observational methods for assessing postures at work. *Applied Ergonomics*, *45*(4), 976–985.

Ding, Z., Li, W., Ogunbona, P., & Qin, L. (2019). A real-time webcam-based method for assessing upper-body postures. *Machine Vision and Applications*, *30*(5), 833–850.

Ding, Z., Wang, P., Ogunbona, P. O., & Li, W. (2017). Investigation of different skeleton features for cnn-based 3D action recognition. In *IEEE international conference on multimedia & expo workshops* (pp. 617–622). IEEE.

Filippeschi, A., Schmitz, N., Miezal, M., Bleser, G., Ruffaldi, E., & Stricker, D. (2017). Survey of motion tracking methods based on inertial sensors: A focus on upper limb human motion. *Sensors*, *17*(6), 1257.

Fu, J., Zheng, H., & Mei, T. (2017). Look closer to see better: Recurrent attention convolutional neural network for fine-grained image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition, vol. 2* (p. 3).

Ghorbel, M., Ammar, S., Kessentini, Y., & Jmaiel, M. (2022). Masking for better discovery: Weakly supervised complementary body regions mining for person re-identification. *Expert Systems with Applications*, *197*, Article 116636. http://dx.doi.org/10.1016/j.eswa.2022.116636, URL https://www.sciencedirect.com/science/article/pii/S0957417422001257.

Guo, H., Wang, G., Chen, X., & Zhang, C. (2017). Towards good practices for deep 3D hand pose estimation. arXiv preprint arXiv:1707.07248.

He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 770–778).

Hignett, S., & McAtamney, L. (2000). Rapid Entire Body Assessment (REBA). *Applied Ergonomics*, *31*(2), 201–205.

Huang, G., Liu, Z., Van Der Maaten, L., & Weinberger, K. Q. (2017). Densely connected convolutional networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, vol. 1, no. 2* (p. 3).

Ionescu, C., Carreira, J., & Sminchisescu, C. (2014). Iterated second-order label sensitive pooling for 3D human pose estimation. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 1661–1668).

Jaderberg, M., Simonyan, K., Zisserman, A., et al. (2015). Spatial transformer networks. In *Advances in neural information processing systems* (pp. 2017–2025).

Karhu, O., Kansi, P., & Kuorinka, I. (1977). Correcting working postures in industry: A practical method for analysis. *Applied Ergonomics*, *8*(4), 199–201.

Kee, D., & Karwowski, W. (2001). LUBA: An assessment technique for postural loading on the upper body based on joint motion discomfort and maximum holding time. *Applied Ergonomics*, *32*(4), 357–366.

Kim, J.-h., Jun, J., & Zhang, B.-t. (2018). Bilinear attention networks. In *NIPS* (pp. 1–13). arXiv:arXiv:1805.07932v2.

Lake, B. M., Ullman, T. D., Tenenbaum, J. B., & Gershman, S. J. (2017). Building machines that learn and think like people. *Behavioral and Brain Sciences*, *40*(April), http://dx.doi.org/10.1017/S0140525X16001837, arXiv:arXiv:1604.00289v1.

Lam, M., Mahasseni, B., & Todorovic, S. (2017). Fine-grained recognition as hsnet search for informative image parts. In *Proceedings of the IEEE conference on computer vision and pattern recognition*.

Fernandez-de Las-Penas, C., Alonso-Blanco, C., Cuadrado, M., & Pareja, J. (2006). Forward head posture and neck mobility in chronic tension-type headache: A blinded, controlled study. *Cephalalgia*, *26*(3), 314–319.

Li, L., & Lin, H.-T. (2007). Ordinal regression by extended binary classification. In *Advances in neural information processing systems* (pp. 865–872).

Li, L., Martin, T., & Xu, X. (2020). A novel vision-based real-time method for evaluating postural risk factors associated with musculoskeletal disorders. *Applied Ergonomics*, *87*, Article 103138.

Li, L., & Xu, X. (2019). A deep learning-based RULA method for working posture assessment. In *Proceedings of the human factors and ergonomics society annual meeting, vol. 63, no. 1* (pp. 1090–1094).

Liu, B., & Ferrari, V. (2017). Active learning for human pose estimation. In *Proceedings of the IEEE international conference on computer vision* (pp. 4363–4372).

Logan, G. D., & Sadler, D. D. (1996). *A computational analysis of the apprehension of spatial relations*. The MIT Press.

Lowe, B., Weir, P., & Andrews, D. (2014). Observation-based posture assessment: Review of current practice and recommendations for improvement. *NOISH*, 32.

Luvizon, D. C., Picard, D., & Tabia, H. (2018). 2D/3D pose estimation and action recognition using multitask deep learning. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 5137–5146).

McAtamney, L., & Corlett, E. N. (1993). RULA: A survey method for the investigation of work-related upper limb disorders. *Applied Ergonomics*, *24*(2), 91–99.

McVeigh, K. H., Murray, P. M., Heckman, M. G., Rawal, B., & Peterson, J. J. (2016). Accuracy and validity of goniometer and visual assessments of angular joint positions of the hand and wrist. *The Journal of Hand Surgery*, *41*(4), e21–e35.

Mehrizi, R., Peng, X., Xu, X., Zhang, S., & Li, K. (2019). A Deep Neural Network-based method for estimation of 3D lifting motions. *Journal of Biomechanics*, *84*, 87–93.

Mehrizi, R., Peng, X., Xu, X., Zhang, S., Metaxas, D., & Li, K. (2018). A Computer Vision Based Method for 3D Posture Estimation of Symmetrical Lifting. *Journal of Biomechanics*.

Mehrizi, R., Xu, X., Zhang, S., Pavlovic, V., Metaxas, D., & Li, K. (2017). Using a marker-less method for estimating L5/S1 moments during symmetrical lifting. *Applied Ergonomics*, *65*, 541–550.

Mellone, S., Palmerini, L., Cappello, A., & Chiari, L. (2011). Hilbert–Huang-based tremor removal to assess postural properties from accelerometers. *IEEE Transactions on Biomedical Engineering*, *58*(6), 1752–1761.

Microsoft (2014). Human interface guidelines v2.0. URL http://download.microsoft.com/download/6/7/6/676611B4-1982-47A4-A42E-4CF84E1095A8/KinectHIG.2.0.pdf.

Muller, A., Pontonnier, C., Robert-Lachaine, X., Dumont, G., & Plamondon, A. (2020). Motion-based prediction of external forces and moments and back loading during manual material handling tasks. *Applied Ergonomics*, *82*.

Mundt, M., Thomsen, W., David, S., Dupré, T., Bamer, F., Potthast, W., et al. (2019). Assessment of the measurement accuracy of inertial sensors during different tasks of daily living. *Journal of Biomechanics*, *84*, 81–86.

Niu, Z., Zhou, M., Wang, L., Gao, X., & Hua, G. (2016). Ordinal regression with multiple output cnn for age estimation. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 4920–4928).

Occhipinti, E., & Colombini, D. (1996). Proposal of a concise index for the evaluation of the exposure to repetitive movements of the upper extremity (OCRA index). *La Medicina del lavoro*, *87*(6), 526–548.

Plantard, P., Shum, H. P., Le Pierres, A.-S., & Multon, F. (2017). Validation of an ergonomic assessment method using Kinect data in real workplace conditions. *Applied Ergonomics*, *65*, 562–569.

Poh, M.-Z., Swenson, N. C., & Picard, R. W. (2010). A wearable sensor for unobtrusive, long-term assessment of electrodermal activity. *IEEE Transactions on Biomedical Engineering*, *57*(5), 1243–1252.

Ray, S. J., & Teizer, J. (2012). Real-time construction worker posture analysis for ergonomics training. *Advanced Engineering Informatics*, *26*(2), 439–455.

Santoro, A., Raposo, D., Barrett, D. G., Malinowski, M., Pascanu, R., Battaglia, P., et al. (2017). A simple neural network module for relational reasoning. In *Advances in neural information processing systems* (pp. 4967–4976).

Sun, Y., Zheng, L., Yang, Y., Tian, Q., & Wang, S. (2018). Beyond part models: Person retrieval with refined part pooling. In *European conference on computer vision*. ECCV.

Wang, X., & Girshick, R. (2018). Non-local neural networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition, vol. 1, no. 3* (p. 4).

Wang, F., Jiang, M., Qian, C., Yang, S., Li, C., Zhang, H., et al. (2017). Residual attention network for image classification. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 3156–3164).

Xu, K., Ba, J., Kiros, R., Cho, K., Courville, A., Salakhudinov, R., et al. (2015). Show, attend and tell: Neural image caption generation with visual attention. In *International conference on machine learning* (pp. 2048–2057).

Youdas, J. W., Carey, J. R., & Garrett, T. R. (1991). Reliability of measurements of cervical spine range of motion—Comparison of three methods. *Physical Therapy*, *71*(2), 98–104.

Zhao, H., Zhang, Y., Liu, S., Shi, J., Loy, C. C., Lin, D., et al. (2018). PSANet: Point-wise spatial attention network for scene parsing. In *European conference on computer vision* (pp. 270–286). Springer.